GREEN REVOLUTION COOLING

WHITE PAPER

A Paradign Shift in Data Center Sustainability

Going Beyond Aisle Containment and Economization

An analysis of how Green Revolution Cooling's technology can help improve the Power, Carbon, and Water Usage Effectiveness (PUE, CUE, and WUE) even at some of the most efficient data centers.

Table of Contents

Introduction	1
Introduction to Oil Submersion Cooling	1
The CarnotJet™ System	2
Energy savings = Lower Carbon and Water Footprint	3
Emissions reduction	4
Water savings	5
Conclusion	6
References	7

Introduction

A few years ago, the industry average PUE was well above 2.0, which meant that organizations spent more on keeping systems cool than hey spent on powering the servers. But over the years, this PUE number has steadily dropped. This drop has primarily been driven by hyperscale operators who have taken prudent steps towards minimizing their environmental impact. The popularity of free air cooling and the increased flexibility of IT equipment to operate at higher temperatures, has facilitated PUEs of as low as 1.15. While it may seem like the low hanging fruit is gone, technologies like liquid immersion cooling promise to bring in the next wave of efficiency improvement. Given the massive scale and exponential growth of data centers, a further reduction to PUEs below 1.05 can have a huge impact on not just the operating costs but on the carbon footprint and on the water requirements for the data centers as well. This paper will look at how GRC's oil immersion cooling technology can help data center operators reduce their energy and water consumption, cost effectively.

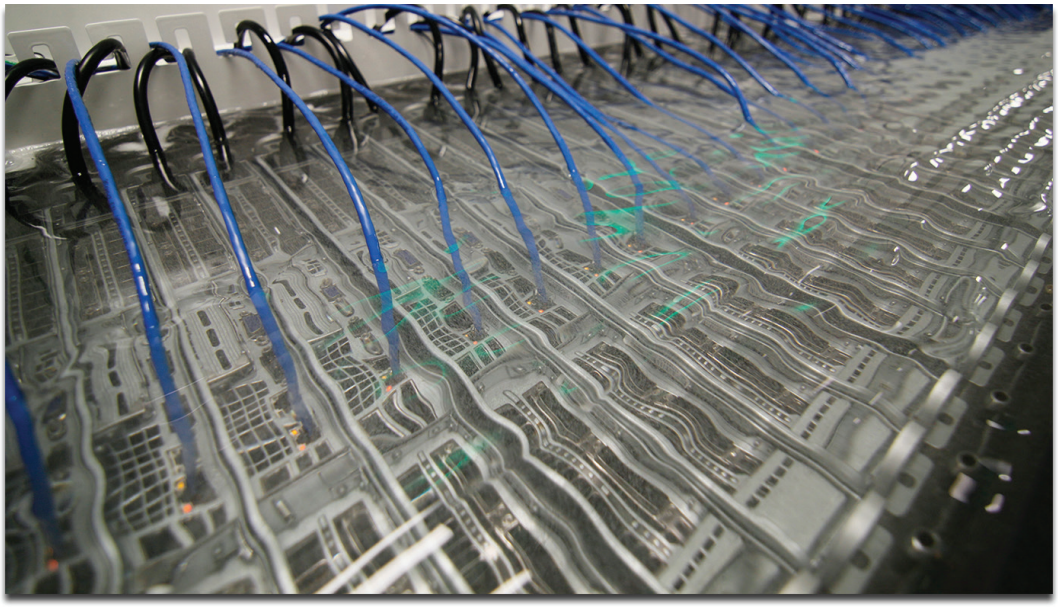
Intro to Oil Immersion Cooling

Oil immersion cooling, as the name suggests is where entire servers are submerged in a mineral oil based dielectric coolant. The ElectroSafe coolant used in GRC's CarnotJet™ system is one such coolant that is a good conductor of heat but not electricity, making it ideal for the cooling of IT equipment. Such methods of cooling have been used for nearly a century in electrical transformers and industrial capacitors. Similar applications of fluid immersion cooling in the world of HPC date back to as early as the 1980s[1]. But concerns regarding the cost, safety, and environmental impact of the coolants used led to limited adoption.

ElectroSafe on the other hand is a non-toxic, eco-friendly, dielectric mineral oil blend. The oils used to make ElectroSafe have a history of being used in not just computing and electric power applications but in medical and domestic applications as well. The high performance, low cost oil is readily available and is rated as 0-1-0 substance by the National Fire Prevention Association (NFPA), meaning that it poses no risk to human health, does not ignite readily, and can be treated with any fire suppression substance, including water[2].

The CarnotJet™ System

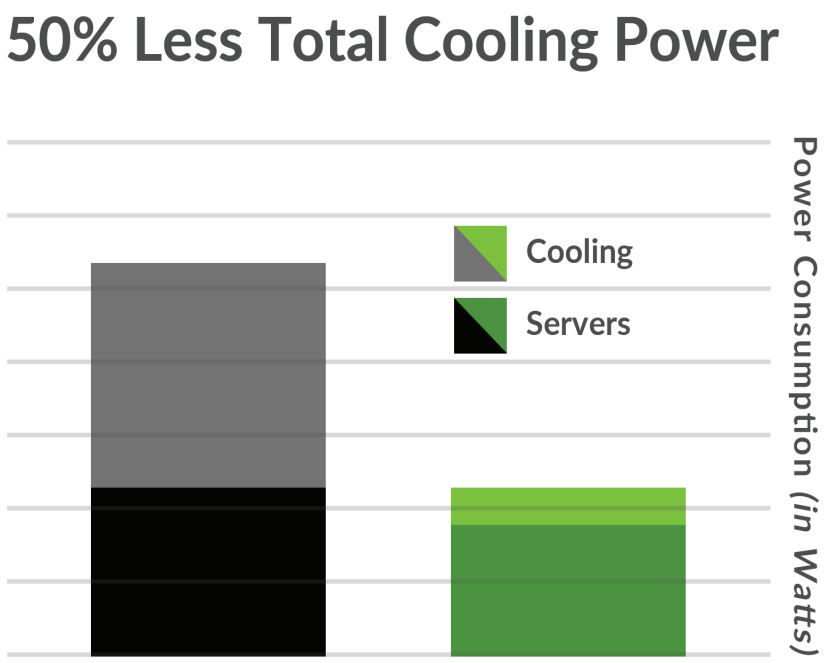
Green Revolution Cooling has brought to the data center market an industrial scale oil submersion cooling solution that works with servers from any manufacturer. The technology consists of four major components; the tank with a horizontal rack (42U), the pump module including the heat exchanger and strainer, the control system, and a cooling tower* to extract heat from the system. The tank is basically a rack that's turned on its back and immersed in a tub of circulating dielectric liquid. The liquid captures heat produced by IT equipment and sends it to a heat exchanger. The open system tanks support servers manufactured by Dell, HP, Supermicro, IBM, Intel, SGI, Quanta, and more.



GRC's CarnotJet server cooling system in action

Energy Savings = Lower Carbon and Water Footprint

The CarnotJet system yields a mechanical PUE of 1.02–1.03 as reported by Intel after a year of testing[3]. This extremely efficient cooling is enabled by the use of the ElectroSafe coolant that has superior heat conductivity and 1,200 times the heat capacity of air. The superior heat conductivity means that ElectroSafe is much more efficient in both extracting heat from the equipment and subsequently expelling it out of the data hall. For example, ElectroSafe at 100°F is as effective at maintaining optimal core temperatures as is air at 70°F. Further, maintaining ElectroSafe at 100°F requires significantly lesser energy than cooling water to 45°F for air cooling.[1]



Furthermore, the CarnotJet system allows reduction in the power consumed by servers as well. This server power reduction is driven by the removal of server fans, and through superior thermal management allowing servers to run cooler. This improvement in energy efficiency directly translates into lower carbon foot print and reduction in the water required for cooling the data center.

**The system can also be used with a dry cooler or other low water consumption alternatives*

Emissions Reduction

On average, one kilowatt-hour of electricity purchased in the United States generates about 1 pound of carbon dioxide[4]. Comparing the carbon footprint of an extremely efficient traditional data center (avg. PUE = 1.21), and a CarnotJet enabled data center (avg. PUE = 1.09) of similar computing capacity reveals an annual Carbon emissions reduction of over 10,000 metric tons per annum, a 21% improvement over what is considered an extremely efficient facility.

	AIR	GRC
Servers	24,000	24,000
Ave Server Power	350 W	280 W
IT Power	8.4 MW	7.392 MW
Cooling Overhead	15%	3%
Electrical Overhead	6%	6%
Effective PUE	1.21	1.09
Total Facility Power	10.16 MW	8.06 MW
Energy Consumption (Yr)	89 Million kWh	70 Million kWh
Carbon Emissions (Yr)	50 Million kgCO2eq	39 Million kgCO2eq
Reduction w/ CarnotJet		21%
Effective CUE	0.68	0.61 kgCO2eq/kWh

CUE or Carbon Usage Effectiveness is a metric defined by Green Grid[5]. CUE is defined as follows

$$\text{CUE} = \frac{\text{Total } CO^2 \text{ Emissions caused by the total Data Center Energy}}{\text{IT Equipment Energy}}$$

The reduction in power directly relates to lower Scope-2 and Scope-1 (from diesel generators) emissions on the CDP (Carbon Disclosure Project), helping organizations achieve better scores on the CDP and reach their carbon reduction goals.

Further, the improved energy efficiency and the resulting decrease in carbon emissions, can give organizations access to energy efficiency grants and programs offered by local utilities, or state and federal governments. PG&E recently published a test report regarding the energy savings with the use of the CarnotJet system and its inclusion in their energy efficiency programs[6].

Water Savings

Apart from the indirect water savings related with the generation of electricity, the CarnotJet system also helps directly reduce the water use and requirements at the data center.

> *"Water is tomorrow's big problem. The water consumption (in data centers) is super embarrassing. It just doesn't feel responsible. We need designs that stop using water."*
>
> -Amazon's James Hamilton

The CarnotJet system enables the use of a hybrid cooling tower or an adiabatic dry cooler, both of which only require additional water when ambient temperature crosses a set point (85° – 95°F). Taking the same example as above, in a data center supporting 24,000 servers the CarnotJet system helps save close to 300 million Liters or 80 million Gallons of water annually, that's close to 90% reduction in water consumption. Certain climates allow for the use of dry coolers in place of cooling towers, thereby reducing water consumption to zero. Examples of such installations include VSC in Vienna, PIC in Barcelona, and a major US Air Force Base.

	Traditional	GRC
Total Facility Power	10.16 MW	8.06 MW
Daily Site Water Usage*	908,400 L	101,270 L
Energy Source Water (Yr)**	160 Million L	127 Million L
Site Water Usage (Yr)	332 Million L	37 Million L
Site Water Savings (Yr)		89%
WUE (Site)	4.51 L/kWh	0.57 L/kWh
WUE (Source)	6.68 L/kWh	2.53 L/kWh

*Based on James Hamilton's estimate[7]
**US average of 1.88 L/kWh
WUE or Water Usage Effectiveness is a metric defined by the Green Grid[8]. It can be defined as follows:

$$\text{WUE} = \frac{\text{Annual Energy Source Water Usage} + \text{Annual Site Water Usage}}{\text{IT Equipment Energy}}$$

Conclusion

Data center energy and water efficiency has improved drastically over the past few years. However, there is still a lot of scope for further improvement. GRC's technology is enabling the next big jump in data center efficiency, and that too cost-effectively.

To learn more about how GRC can help you cut Carbon emissions, and water consumption while lowering build, and operating costs mail us at info@grcooling.com.

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